

Group Project — Building an AI-Supported FSA Agent

Over the 24 sessions of this course each group builds, in four blocks of six sessions, an AI assistant that supports financial statement analysis, and uses it to produce an equity-research note on one listed company. The agent is developed incrementally: each block adds the analytical skills covered in class during that block.

You are a group of analysts; the agent you are building is a junior analyst: fast, tireless, extremely well read, and every so often confidently wrong. Nothing in these instructions is designed to make it produce the right answer on its own — the point of this group project is to make you the senior analysts who know precisely when, and why, not to believe it

1. What you deliver

- The agent. A documented, versioned prompt architecture (v1.0 → v4.0) together with its supporting calculation workbook.
- The FSA report. 8–10 pages (word document) in the format of a sell-side equity note: a front-page recommendation box (Buy / Hold / Sell, target price, key risks), then the analysis, and — on the final page — a mandatory section, *The limits of this agent*, which has to be written after Session 24. Annexes are not counted and are not limited.
- Three progress notes. One page each, submitted at the end of Blocks 1, 2 and 3.

Deadlines. The agent and the report are *not* due together. Agent v4.0 (see below) is final and frozen on Thursday 3 December, 23:59 (night before the last class); the report follows on Thursday 22 December, 23:59 (night after the exam).

2. Your company

Each group must choose one of the following companies. The 2025 Annual Report and other relevant documents are on Blackboard.

Sector	Company assigned (Annual Report 2025)	Country
Defence	Rheinmetall AG	Germany
Fashion & Luxury	LVMH Moët Hennessy Louis Vuitton	France
Telecommunications	Telefónica S.A.	Spain
Pharmaceuticals	Novo Nordisk A/S	Denmark
Semiconductors	STMicroelectronics N.V.	France–Italy

3. What we mean by “agent”

Prompt architecture (mandatory). A *master prompt* that fixes the agent’s role, the inputs it requires, the reporting framework it applies, the sourcing rules it must obey and the format of its output; plus one *module prompt* per block. Build it in a persistent workspace (ChatGPT Project or Custom GPT, Claude Project, Copilot notebook, Gemini Notebook) with the annual report and your workbook loaded as knowledge. Concretely, the master prompt is the permanent instruction set of that workspace — the Project instructions in ChatGPT or Claude, the Instructions field of a Custom GPT — and not the opening message of a chat, which disappears with the conversation; each module prompt is the task you give the agent for that block, kept in the same workspace as a separate, named file. No programming is required, and building a tool-using agent or an API pipeline earns no credit in itself: what we assess is the specification and the analysis. Taken together, your master prompt and the module in force must cover all eight of the following. Components 1, 3, 6 and 8 belong in the master prompt, where they stay constant; components 2, 4, 5 and 7 belong in each module prompt, because they change with the task.

WHAT THE PROMPT ARCHITECTURE MUST CONTAIN

1. **Role and mandate.** Who the agent is, which decision its output serves, and what it must not attempt.
2. **Declared inputs.** Which documents, which years, which sections — and an instruction to stop and ask when an input is missing, rather than proceed on an assumption.
3. **Definitions and conventions.** Every term the agent must not improvise: core operating income, the definition of debt, average or ending invested capital, currency, units, rounding. What you leave undefined the model will decide — differently each time.
4. **The procedure, numbered.** One operation per step, in the order an analyst performs them. Mark which figures the agent must take from the workbook and must not recompute, and which steps it performs itself.
5. **Thresholds and decision rules.** The numeric triggers that turn an observation into a flag — interest coverage below x, net debt/EBITDA above y, an earnings-to-cash gap wider than z% of revenue. You set them and justify them for your sector; the agent reports the value found beside the threshold.
6. **Evidence and sourcing.** Every figure with its document and page or note; every external claim with a verbatim quotation and a URL; anything not found reported as not found, never quietly estimated.
7. **Prescribed output.** The exact sections and tables, with their column headings; the length; and a closing statement of what could not be verified.
8. **Self-check.** A final step listing the assumptions made, the figures it could not source, and any inconsistency noticed — flagged, never smoothed over.

Order of magnitude: a module prompt that meets these eight requirements runs to one or two pages of text. “Analyse the balance sheet and compute the main ratios” is not a prompt, it is a wish. **The test to apply before every checkpoint:** give your module prompt to a group working in another sector. If they can run it on *their* company without asking you a single question, it is specified. If they cannot, it is not.

How the architecture evolves. The master prompt is not frozen at v1.0: as the agent reveals what you failed to define, you extend it — a definition of net financial position in v2.0, a sourcing rule in v3.0. The module prompts accumulate rather than replace one another. At every checkpoint you therefore submit a single file containing the current master prompt and every module written so far, preceded by a change note of one line per version: “v2.0 — added the definition of net financial position, after the agent used three different ones in the same output”.

Calculation workbook (strongly recommended). An Excel file in which the reformulations and the ratios are actually computed. The workbook is yours: you build it, with your own formulas, and you own every number in it. The agent reads it, checks it for internal consistency and interprets it, but it never writes to it — any correction it proposes is one you make yourselves. It is not asked to do arithmetic in prose. Groups that leave the computation to the model will spend Block 3 discovering why this matters.

One design rule from Session 1: your agent must be company-agnostic. No figure taken from your company may be hard-coded into a prompt. A well-built agent can be pointed at the annual report of any company in your sector and still do its job; a prompt that only works because you wrote your own numbers into it is not an agent, it is a transcript.

4. Five important rules

1. Every figure in the report is traceable to a primary source. State the document and the page or note.
2. The agent may not cite what it has not been given or has not retrieved with a verifiable URL. A fabricated reference voids the section that rests on it.
3. **Keep an Agent Log**, on the template posted on Blackboard, one row per entry: date | agent version | what you asked | what the agent produced that was wrong, missing or oversimplified (quoted) | how you established that it was wrong (source, page or note) | what you changed, or why you changed nothing. An entry is one defect and its resolution, not one prompt: a conversation of twenty exchanges that revealed two errors is two entries. Minimum five entries per block. It is submitted at every checkpoint and it is assessed.
4. Prompts are submitted in full, in plain text (.md or .txt). We will read them.
5. You are accountable for the report. “The agent said so” is not a defence, in this course or anywhere else.

5. The four blocks

Block	What the agent module should be able to do	Checkpoint deliverables
BLOCK 1 The Reformulation Engine Sessions 1–6 3 – 18 September Chapters 1–7	Module A — from the reported statements to analytical ones. The agent should return, for three years: • the reformulated income statement (core vs. non-core operating income; unusual items identified, quantified and justified); • the operating cost structure under the three lenses (gross profit; added value and EBITDA; contribution margin); • the managerial balance sheet (invested capital, net financial position, operating liabilities); • the ROI and ROE decompositions (DuPont, Modigliani-Miller, Penman) with a written reading of what drives each.	Agent v1.0 (master prompt + Module A) Reformulated IS, managerial BS, ratio table (workbook) Agent Log, Block 1 Progress note 1 Due: Sun 27 September, 23:59
BLOCK 2 Cash and the Business Model Sessions 7–12 24 September – 9 October Chapters 8–12	Module B — does the cash confirm the profit? • reformulate the IAS 7 / IFRS 18 cash flow statement into the OFCF and FCFE structure; • compute the cash-flow ratios and read the wedge between earnings and operating cash flow; • screen for creative cash-flow reporting (classification shifting, acquisition-driven flows, receivables sales, capitalised costs) and estimate sustainable operating cash flow; • describe the business model and translate it into 5–8 industry KPIs, benchmarked against two or three sector peers you identify.	Agent v2.0 (Modules A–B) Reformulated cash flow statement + cash-flow ratios KPI and peer-benchmark table Agent Log, Block 2 Progress note 2 Due: Sun 18 October, 23:59
BLOCK 3 Quality of the Reported Numbers Sessions 13–18 16 October – 13 November Chapters 13–16	Module C — can these numbers be trusted? • reconcile management’s non-GAAP measures and street earnings to the IFRS figures, and classify every adjustment as informative or aggressive; • assess intangibles and goodwill (recognition, internally generated assets, impairment testing and its assumptions); • identify where an accounting policy choice, rather than economics, is driving a ratio, and quantify the effect; • run a red-flag scan on accounting quality (accruals, revenue recognition, capitalisation, related parties, auditor’s key audit matters) and output a rated table: evidence → implication → severity.	Agent v3.0 (Modules A–C) Non-GAAP / street-earnings reconciliation Rated red-flag table Agent Log, Block 3 Progress note 3 Due: Sun 22 November, 23:59

Block	What the agent module should be able to do	Checkpoint deliverables
BLOCK 4 From Analysis to a Recommendation Sessions 19–24 19 November – 4 December <i>Chapters 17–18</i>	Module D — what is it worth, and what would you do? • build a five-year forecast (seven-step game plan) in which every assumption is explicitly derived from the findings of Blocks 1–3; • value the equity by DCF and by residual income, and cross-check against market multiples; • run sensitivity and scenario analysis on the value drivers (Monte Carlo simulation optional); • state a Buy / Hold / Sell recommendation with target price, the three risks that most threaten it, and the evidence that would make you change your mind.	Agent v4.0 — final and frozen Complete Agent Log and prompt version history Due: Thu 3 December, 23:59 — then, separately — Session 24 raw output (see §6) Due: Fri 4 December, 23:59 Final FSA report (8–10 pp.) + Analyst’s Verdict Due: Thu 22 December, 23:59

6. Session 24 on our last day of class — the agent goes to work...

The last session is your agent’s field test. Bring v4.0 *working*, on a laptop you can run it from; the financial statements and other relevant documents to be analysed are released at the start of the session. Prompt editing during the run is not allowed — the agent under test is the version you submitted on 3 December (and we hold that file). Write down what you would have changed instead: that note is worth more to us than the change.

Two outcomes for this session:

- **The raw output.** The agent’s unedited output, together with an export of the chat, uploaded by 23:59 on Friday 4 December. Do not clean it, do not fix it, do not re-run it selectively. We want to see what it actually produced.
- **The Analyst’s Verdict.** One page, on the template below, submitted with the final report on 22 December.

ANALYST’S VERDICT

1. The recommendation the agent produced (Buy / Hold / Sell, target price if any) — one line.
2. Three things it got right, and the evidence that convinced you.
3. Three things it got wrong, missed, or invented — quote its own words.
4. The parts of its output you would refuse to sign, and why.
5. What you would change in the prompt — and what no prompt could have fixed.

Three documents, easily confused. The Session 24 raw output is what the agent produced in class on a company you have never analysed: unedited, and no part of it belongs in the report. The Analyst’s Verdict is your one-page judgement on that output, submitted as a separate file. And *The limits of this agent* is the closing page of the FSA report: it concerns your own company and the whole semester, informed by Session 24 but not a summary of it — where the agent proved unreliable, which judgements no prompt could delegate, and what you would rebuild if you started again. You will receive our feedback on the raw output and on the Verdict together with the grade.

7. Progress notes

One page, submitted together with the deliverables of Blocks 1, 2 and 3. Not a summary of what you did — the files already show that. Each note answers three questions: (i) what does your agent do now that it could not do six sessions ago? (ii) which methodological choice was the hardest, and why did you decide as you did? (iii) where did the agent mislead you, and how did you catch it? Write it yourselves: a page written by the agent about the agent tells us nothing, and it is recognisable at a glance. You will receive timely feedback from us.

8. Submission

Blackboard, folder *Group project*, by 23:59 on the stated date. Use exactly this naming convention, where *nn* is your group number and *k* the block:

Gnn_Bk_Agent_vX.md · Gnn_Bk_Workbook.xlsx · Gnn_Bk_Output.pdf · Gnn_Bk_Log.xlsx · Gnn_Bk_Note.pdf

For the last session and the report: **Gnn_S24_RawOutput.pdf · Gnn_S24_ChatExport.pdf · Gnn_S24_Verdict.pdf · Gnn_FSA_Report.pdf**

9. How the project is assessed (tentative guidelines)

Six dimensions, in this order of importance: the technical correctness of the reformulations, ratios and valuation; the specificity and portability of the prompt architecture; how the agent performs in the Session 24 field test; the visible evidence of iteration from v1.0 to v4.0; the critical use of AI documented in the Agent Log and the Analyst’s Verdict; and the clarity and defensibility of the final recommendation.